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CUET UG

Chemistry

Haloalkanes and Haloarenes

Official Syllabus Unit VI · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus, the current 2027 baseline, Unit VI)	Skills the questions test
Classification & C–X Nature	Classification of haloalkanes and haloarenes (1°/2°/3°, allylic, benzylic, vinylic, aryl), systematic IUPAC nomenclature, nature of the polar C–X bond.	Assigning IUPAC names, classifying alkyl/aryl halides, predicting dipole moments, bond lengths, and C–X bond dissociation enthalpies.
Methods of Preparation	Synthesis from "alc"ohols (HX, PX_3, PCl_5, $SOCl_2$), hydrocarbons (free-radical halogenation, electrophilic addition/substitution), halide exchange (Finkelstein, Swarts), and diazonium salts (Sandmeyer, Balz-Schiemann).	Selecting optimal reagents (e.g. Darzens thionyl chloride route), applying Markovnikov vs anti-Markovnikov peroxide addition, and preparing fluoro/iodoalkanes.
Physical & Chemical Properties	Melting & boiling points, density, water solubility. Nucleophilic substitutions (S_N1 and S_N2), stereochemical aspects (optical activity, chirality, retention, inversion, racemisation).	Contrasting S_N1 vs S_N2 rate laws, identifying substrate reactivity hierarchies ($3^\circ > 2^\circ > 1^\circ$ vs $1^\circ > 2^\circ > 3^\circ$), and predicting stereochemical outcomes (Walden inversion vs racemisation).
Eliminations & Organometallics	β-elimination (dehydrohalogenation, Saytzeff rule), reactions with metals (Grignard reagent formation and hydrolysis, Wurtz, Fittig, and Wurtz-Fittig couplings).	Predicting major alkene products in dehydrohalogenation, designing Grignard syntheses with active hydrogen substrates, and c"alc"ulating coupling yields.
Reactivity of Haloarenes	Inertness of aryl halides towards nucleophilic substitution, Dow's process, activating influence of electron-withdrawing groups ($-NO_2$ at ortho/para), electrophilic substitution reactions.	Explaining resonance partial double-bond character, rationalizing S_NAr acceleration by ortho/para nitro substituents, and deducing major electrophilic substitution isomers.

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus, the current 2027 baseline, Unit VI)	Skills the questions test
Polyhalogen Compounds	Uses and environmental impact of dichloromethane (CH_2Cl_2), chloroform ($CHCl_3$), iodoform (CHI_3), carbon tetrachloride (CCl_4), freons (CFCs), and DDT.	Detailing phosgene formation, antiseptic action of iodoform, ozone layer destruction mechanisms by CFCs, and ecological biomagnification of DDT.

EXAM PATTERN

CUET UG Paper Pattern & Weightage: The Chemistry paper carries **50 compulsory questions** in **60 minutes** (+5 marks for correct, −1 mark for incorrect). Unit VI (Haloalkanes and Haloarenes) opens the Class 12 organic syllabus and contributes 4 to 6 questions focusing on S_N1 vs S_N2 stereochemistry, Saytzeff dehydrohalogenation, ambidentate nucleophiles, Dow's process activation, and polyhalogen environmental impacts.

Module 1: Classification, Nomenclature & Nature of the C–X Bond

1. Structural Classification of Halogen Derivatives

Halogen compounds are classified based on the number of halogen atoms and the hybridisation state of the carbon atom bearing the halogen:

1. **By Halogen Count:** Monohaloalkanes (CH_3 Cl), Dihaloalkanes (CH_2Cl_2, 1,2-dichloroethane), and Polyhaloalkanes ($CHCl_3$, CCl_4).
   - **Geminal Dihalides (gem-dihalides):** Both halogen atoms are attached to the same carbon atom (e.g. ethylidene chloride, CH_3 CH Cl_2).
   - **Vicinal Dihalides (vic-dihalides):** Halogen atoms are attached to adjacent carbon atoms (e.g. ethylene dichloride, Cl $CH_2 - CH_2$ Cl).
2. **Compounds Containing sp^3 Carbon-Halogen ($C(sp^3) - X$) Bonds:**
   - **Alkyl Halides (Haloalkanes):** Halogen atom is bonded to an alkyl chain ($R - X$). Subdivided into primary (1°), secondary (2°), and tertiary (3°) based on the degree of substitution of the α-carbon.
   - **Allylic Halides:** Halogen atom is bonded to an sp^3-hybridised carbon atom adjacent to a carbon-carbon double bond ($CH_2 = CH - CH_2X$).
   - **Benzylic Halides:** Halogen atom is bonded to an sp^3-hybridised carbon atom directly attached to an aromatic benzene ring ($C_6H_5CH_2X$).
3. **Compounds Containing sp^2 Carbon-Halogen ($C(sp^2) - X$) Bonds:**
   - **Vinylic Halides:** Halogen atom is bonded directly to one of the carbon atoms of a double bond ($CH_2 = CH - X$).
   - **Aryl Halides (Haloarenes):** Halogen atom is bonded directly to an aromatic ring carbon ($C_6H_5 - X$).

2. Nature of the Polar C–X Bond

Because halogens possess substantially higher electronegativity than carbon ($F = 4.0, \text{Cl} = 3.0, \text{Br} = 2.8, I = 2.5$), the carbon-halogen bond is intensely polar with partial charges: $C^{\delta+} - X^{\delta-}$.

- **Bond Length:** Increases steadily down Group 17 as atomic radius expands:

$$C - F(139 \text{ pm}) < C - \text{Cl } (178 \text{ pm}) < C - \text{Br } (193 \text{ pm}) < C - I(214 \text{ pm})$$

- **Bond Dissociation Enthalpy:** Decreases sharply down the group as orbital overlap efficiency diminishes:

$$C - F(452\text{kJ mol}^{-1}) > C - \text{Cl } (351) > C - \text{Br } (293) > C - I(234)$$

  CUET Direct Deduction: The weak $C - I$ bond makes alkyl iodides the most reactive towards substitution and elimination.

- **Dipole Moment (μ):** Dipole moment is the product of charge separation and bond length ($\mu = q \times d$).

$$\text{CH}_3 \text{ Cl } (1.860 \text{ D}) > \text{CH}_3F(1.847 \text{ D}) > \text{CH}_3 \text{ Br } (1.830 \text{ D}) > \text{CH}_3I(1.636 \text{ D})$$

  Critical Examination Trap: Although fluorine has the highest electronegativity, the $C - F$ bond distance is extremely short. Consequently, the longer bond distance in chloromethane (CH_3 Cl) overcomes the slight charge difference, giving CH_3 Cl a larger dipole moment than CH_3F.

Haloalkane	Bond Length (pm)	Enthalpy (kJ/mol)	Dipole Moment (μ, D)
$\text{CH}_3 - F$	139	452	1.847
$\text{CH}_3 - \text{Cl}$	178	351	1.860 (Maximum)
$\text{CH}_3 - \text{Br}$	193	293	1.830
$\text{CH}_3 - I$	214	234 (Minimum)	1.636

Module 2: Methods of Preparation of Haloalkanes & Haloarenes

1. Synthesis of Haloalkanes from Alcohols

Alcohols are the most versatile starting precursors for preparing alkyl halides via replacement of the hydroxyl (−OH) group:

1. **Reaction with Hydrogen Halides (HX):**

$$R - \text{OH} + \text{HX} \xrightarrow{\text{anhy. } \text{ZnCl}_2} R - X + \text{H}_2\text{O}$$

   - **Reactivity of Alcohols:** $3^\circ > 2^\circ > 1^\circ$ (governed by the stability of carbocation intermediates).
   - **Reactivity of Halogen Acids:** HI > HBr > HCl > HF.
   - **Lucas Reagent (Conc. HCl + ZnCl_2):** Distinguishes "alc"ohol classes based on the appearance of insoluble alkyl chloride cloudiness/turbidity:
     - 3° "alc"ohols: Instant turbidity at room temperature.
     - 2° "alc"ohols: Turbidity appears within 5 minutes.
     - 1° "alc"ohols: Turbidity appears only upon heating.

2. **Reaction with Phosphorus Halides (PCl_5 and PCl_3):**

$$R - \text{OH} + \text{PCl}_5 \rightarrow R - \text{Cl} + \text{POCl}_3 + \text{HCl}$$

$$3R-\mathrm{OH}+\mathrm{PCl_3}\rightarrow 3R-\mathrm{Cl}+\mathrm{H_3PO_3}$$

Alkyl bromides and iodides are prepared similarly using phosphorus tribromide ($\mathrm{PBr_3}$) or triiodide ($\mathrm{PI_3}$) generated in situ from red phosphorus and $\mathrm{Br_2}$ or I_2.

3. **Reaction with Thionyl Chloride ($\mathrm{SOCl_2}$ — Darzens Process):**

$$R-\mathrm{OH}+\mathrm{SOCl_2}\xrightarrow{\text{Pyridine}}R-\mathrm{Cl}+\mathrm{SO_2}\uparrow+\mathrm{HCl}\uparrow$$

Preferred Industrial & Laboratory Method: Because both byproducts—sulfur dioxide ($\mathrm{SO_2}$) and hydrogen chloride (HCl)—are gases that escape from the reaction mixture, the reaction yields pure alkyl chloride without requiring complex purification steps.

2. Synthesis from Hydrocarbons

1. **Free Radical Halogenation of Alkanes:** Chlorination or bromination under ultraviolet light or heat gives complex isomeric mixtures of mono-, di-, and polyhaloalkanes:

$$\mathrm{CH_4}+\mathrm{Cl_2}\xrightarrow{h\nu}\mathrm{CH_3\ Cl}+\mathrm{CH_2Cl_2}+\mathrm{CHCl_3}+\mathrm{CCl_4}$$

2. **Electrophilic Addition to Alkenes:**
   - **Addition of Hydrogen Halides (HX):** Follows **Markovnikov's Rule**: the negative part of the addendum (X^-) attaches to the carbon atom holding the fewer hydrogen atoms.

$$\mathrm{CH_3-CH=CH_2}+\mathrm{HBr}\rightarrow\mathrm{CH_3-CH(Br)-CH_3}\quad\text{(Major: 2-bromopropane)}$$

   - **"Peroxide" Effect (Kharasch Effect):** In the presence of organic peroxides (e.g. benzoyl peroxide), the addition of HBr (and **only** HBr) occurs contrary to Markovnikov's rule:

$$\mathrm{CH_3-CH=CH_2}+\mathrm{HBr}\xrightarrow{\text{Peroxide}}\mathrm{CH_3-CH_2-CH_2\ Br}\quad\text{(Major: 1-bromopropane)}$$

CUET Mechanistic Fact: "Peroxide" effect does not operate with HCl or HI because the H-Cl bond is too strong (430.5 kJ/mol) for free radical cleavage, while the $C-I$ addition step is endothermic.

   - **Addition of Halogens:** Addition of $\mathrm{Br_2}$ in $\mathrm{CCl_4}$ across a double bond discharges the characteristic reddish-brown bromine colour (**test for unsaturation**).

3. Halide Exchange Reactions

Specialised routes to synthesise alkyl iodides and fluorides that are inaccessible via direct halogenation:

1. **Finkelstein Reaction (Synthesis of Alkyl Iodides):** Alkyl chlorides or bromides are treated with sodium iodide (NaI) in "dry acetone":

$$R-X+\mathrm{NaI}\xrightarrow{\text{Dry Acetone}}R-I+\mathrm{NaX}\downarrow\quad(X=\mathrm{Cl,Br})$$

Thermodynamic Driving Force: Sodium chloride (NaCl) and sodium bromide (NaBr) are insoluble in "dry acetone" and precipitate out, shifting the equilibrium forward according to Le Chatelier's principle.

2. **Swarts Reaction (Synthesis of Alkyl Fluorides):** Heating alkyl chlorides or bromides with heavy metal fluorides such as silver fluoride (AgF), mercurous fluoride ($\mathrm{Hg_2F_2}$), cobalt trifluoride ($\mathrm{CoF_3}$), or antimony trifluoride ($\mathrm{SbF_3}$):

$$\mathrm{CH_3-Br}+\mathrm{AgF}\rightarrow\mathrm{CH_3}-F+\mathrm{AgBr}\downarrow$$

4. Synthesis of Haloarenes from Diazonium Salts

1. **Sandmeyer Reaction:** Freshly prepared benzenediazonium chloride is treated with cuprous chloride or cuprous bromide dissolved in the corresponding halogen acid:

$$Ar - N_2^+Cl^- + Cu_2\frac{Cl_2}{HCl} \rightarrow Ar - Cl + N_2 \uparrow$$

$$Ar - N_2^+Cl^- + Cu_2\frac{Br_2}{HBr} \rightarrow Ar - Br + N_2 \uparrow$$

2. **Gattermann Reaction:** Uses copper powder in the presence of HCl or HBr (lower yield than Sandmeyer).
3. **Preparation of Iodoarenes:** Benzenediazonium chloride is simply warmed with aqueous potassium iodide (KI):

$$Ar - N_2^+Cl^- + KI \rightarrow Ar - I + KCl + N_2 \uparrow \quad \text{(No cuprous catalyst required)}$$

4. **Balz-Schiemann Reaction (Fluoroarenes):** Benzenediazonium chloride is treated with fluoroboric acid (HBF_4) to precipitate benzenediazonium fluoroborate, which decomposes upon heating:

$$Ar - N_2^+Cl^- + HBF_4 \rightarrow Ar - N_2^+BF_4^- \xrightarrow{\Delta} Ar - F + BF_3 \uparrow + N_2 \uparrow$$

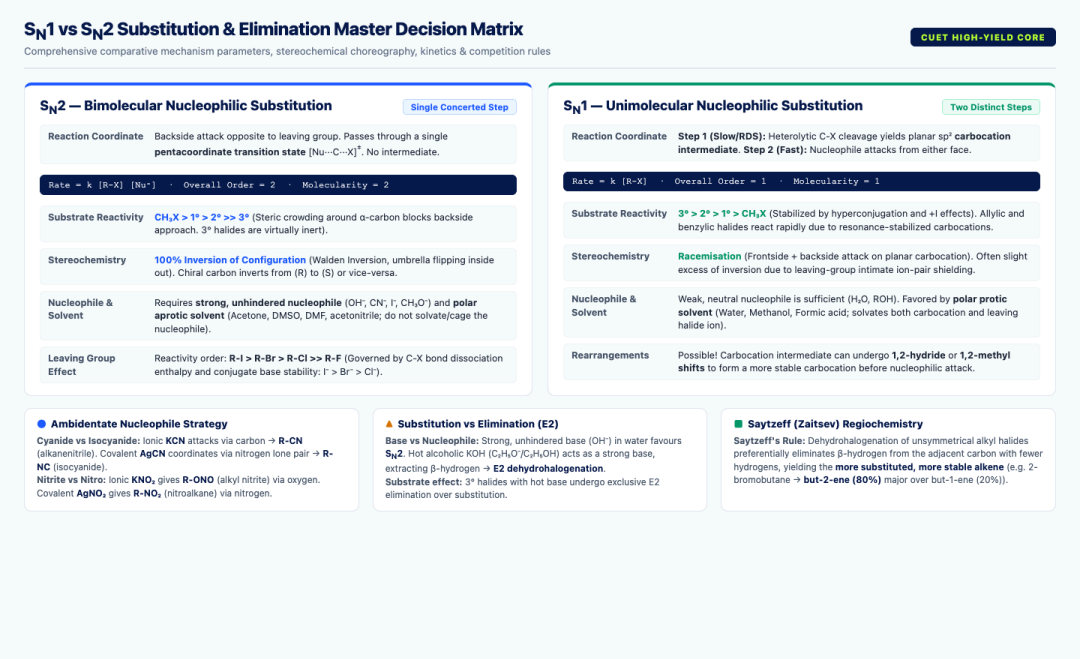


Figure 1: Bimolecular (S_N2) vs Unimolecular (S_N1) nucleophilic substitution comparison matrix: reaction coordinates, rate laws, stereochemical inversion vs racemisation, substrate orders, and Saytzeff elimination rules.

Module 3: Physical Properties & Substitution Mechanisms (S_N1 vs S_N2)

1. Physical Properties Trends

- **Boiling Points:** Alkyl halides have higher boiling points than parent alkanes due to greater polarity and stronger dipole-dipole attractions.
  - For a given alkyl group, boiling points increase down the halogen group:

$$R-I > R-\mathrm{Br} > R-\mathrm{Cl} > R-F$$

- Among isomeric alkyl halides, boiling point decreases with increasing branching:

  n-butyl bromide (375 K) > isobutyl bromide (364 K) > tert-butyl bromide (346 K)

  Reason: Branching makes the molecule more spherical, diminishing surface area and weakening London dispersion forces.

- **Melting Points of Dihalobenzenes:** The **para-isomer** possesses a significantly higher melting point than the ortho- and meta-isomers:
  - **p-Dichlorobenzene** melts at 325 K, whereas the ortho- (256 K) and meta- (249 K) isomers melt substantially lower.
  - **Symmetry Factor:** The symmetrical para-isomer packs tightly and uniformly into the crystalline lattice, requiring higher thermal energy to break.
- **Density & Solubility:** Density follows $R-I > R-\mathrm{Br} > R-\mathrm{Cl}$. Polyhaloalkanes and bromo/iodoalkanes are denser than water. Alkyl halides are practically insoluble in water because the energy released from dipole-dipole attractions between haloalkane and water is insufficient to break existing hydrogen bonds between water molecules.

2. Bimolecular Nucleophilic Substitution (S_N2)

The reaction proceeds in a single, concerted step without any intermediates:

1. **Mechanism & Transition State:** The nucleophile attacks the α-carbon from the backside opposite to the leaving halide ion (180°). A pentacoordinate transition state $[\mathrm{N}\cdots C\cdots X]^{\ddagger}$ forms, in which carbon is simultaneously bonded to both groups.
2. **Kinetics:** The rate depends on the concentrations of both the alkyl halide and the nucleophile:

   $$\text{Rate} = k[R-X][\mathrm{N}^-] \quad \text{(Second order overall, bimolecular)}$$

3. **Stereochemistry:** Yields complete **100% Inversion of Configuration** (**Walden Inversion**), analogous to an umbrella flipping inside out in strong wind. If the starting substrate is chiral (e.g. $(-)-2-$bromooctane), it converts into the pure opposite enantiomer $(+)-2-$octanol.
4. **Substrate Reactivity Hierarchy:** Steric hindrance around the α-carbon dictates reactivity:

   $$\mathrm{CH_3}X > 1^{\circ}(1^{o}) > 2^{\circ}(2^{o}) \gg 3^{\circ}(3^{o})$$

   Tertiary alkyl halides are virtually unreactive towards S_N2 because bulky alkyl groups completely shield the backside of the carbon.
5. **Optimal Reaction Conditions:** High concentration of strong, unhindered nucleophile ($\mathrm{OH^-}, \mathrm{CN^-}, \mathrm{CH_3O^-}$) in a **polar aprotic solvent** (acetone, DMSO, DMF).

3. Unimolecular Nucleophilic Substitution (S_N1)

The reaction proceeds in two distinct sequential steps through a carbocation intermediate:

1. **Mechanism:**
   - **Step 1 (Slow, Rate-Determining Step):** Heterolytic cleavage of the $C-X$ bond generates a planar, sp^2-hybridised carbocation intermediate and a halide ion:

     $$R-X \overset{\text{slow}}{\rightarrow} R^+ + X^-$$

   - **Step 2 (Fast):** The nucleophile attacks the planar carbocation from either the frontside or the backside with equal statistical probability:

     $$R^+ + \mathrm{N}^- \overset{\text{fast}}{\rightarrow} R-\mathrm{N}$$

2. **Kinetics:** Nucleophile does not participate in the rate-determining step:

$$\text{Rate} = k[R-X] \quad \text{(First order overall, unimolecular)}$$

3. **Stereochemistry:** Produces **Racemisation** (formation of a 50:50 mixture of dextrorotatory and laevorotatory enantiomers, resulting in net optical inactivity). In practice, a slight excess of inversion is observed because the departing halide ion lingers briefly near the front face, partially shielding it (**intimate ion-pair effect**).
4. **Substrate Reactivity Hierarchy:** Governed by carbocation stability (hyperconjugation and $+I$ inductive effects):

$$3^{\circ} > 2^{\circ} > 1^{\circ} > CH_3X$$

   - **Allylic and Benzylic Halides:** Exhibit extraordinary S_N1 reactivity because the resulting carbocations are stabilized by resonance delocalisation across the adjacent π-system.
5. **Optimal Conditions:** Weak, neutral nucleophile (H_2 O, ROH) in a **polar protic solvent** (water, "alc"ohols, formic acid) that stabilizes ions via hydrogen bonding.

4. Ambidentate Nucleophiles & Product Divergence

Nucleophiles possessing two distinct donor atoms through either of which they can coordinate:

1. **Cyanide Ion (CN^-):**
   - **Reaction with KCN (Ionic):** Forms alkyl cyanides / nitriles ($R-CN$). Because the $C-C$ bond is more thermodynamically stable (347 kJ/mol) than the $C-N$ bond (305 kJ/mol), attack occurs predominantly through carbon.
   - **Reaction with AgCN (Covalent):** Forms alkyl isocyanides / carbylamines ($R-NC$). Silver-carbon bond has high covalent character; only the nitrogen lone pair is available for nucleophilic donation.
2. **Nitrite Ion (NO_2^-):**
   - **Reaction with KNO_2 (Ionic):** Forms alkyl nitrites ($R-O-N=O$) via attack through oxygen.
   - **Reaction with $AgNO_2$ (Covalent):** Forms nitroalkanes ($R-NO_2$) via attack through nitrogen.

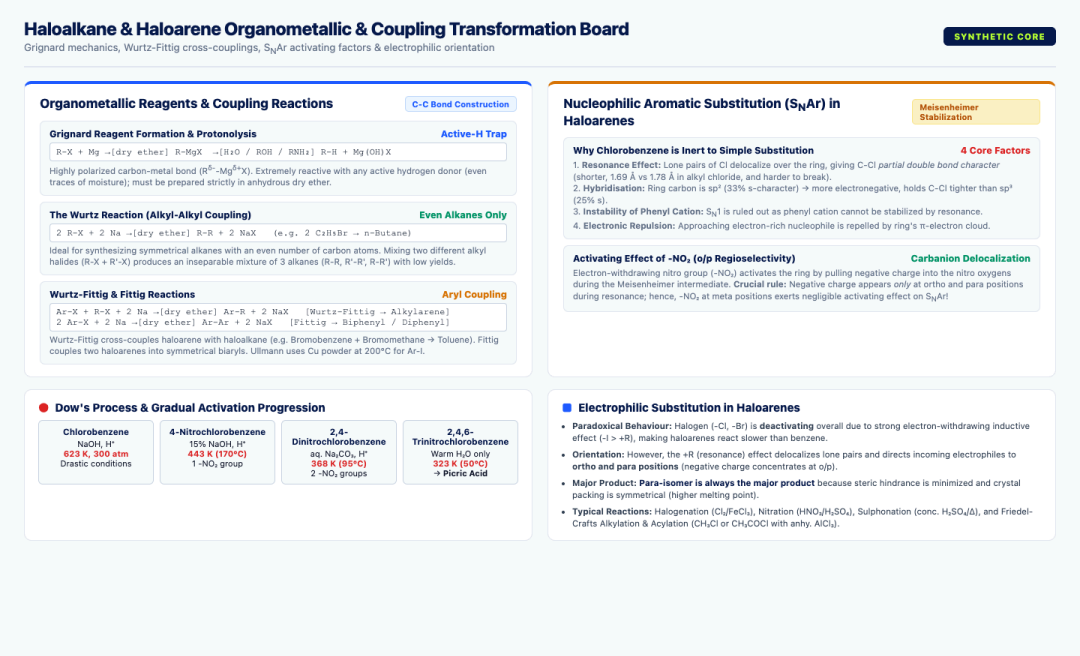


Figure 2: Haloalkane & haloarene organometallic transformations: Grignard reagent hydrolysis, Wurtz-Fittig cross-couplings, Dow's process S_NAr activation, and electrophilic substitution regiochemistry.

Module 4: Elimination Reactions & Organometallic Couplings

1. Dehydrohalogenation (β-Elimination) & Saytzeff's Rule

When a haloalkane containing at least one β-hydrogen is heated with "alc"oholic potassium hydroxide (alc. KOH), hydrogen and halogen are eliminated from adjacent carbons, generating an alkene:

$$CH_3 - CH_2 - CH(Br) - CH_3 + KOH\ (alc) \xrightarrow{\Delta} CH_3 - CH = CH - CH_3 (81\%) + CH_3 - CH_2 - CH = CH_2 (19\%)$$

- **Saytzeff's Rule (Zaitsev's Rule):** In dehydrohalogenation reactions, the preferred major product is that alkene which has the greater number of alkyl groups attached to the doubly bonded carbon atoms (**more substituted, thermodynamically more stable alkene**).
- **Hofmann Elimination (Minor Exception):** When dehydrohalogenation is conducted using a very bulky base (e.g. potassium tert-butoxide, $t - BuO^- K^+$) or when the leaving group is fluoride ($R - F$), the steric crowding causes base to abstract the more accessible β-hydrogen from the least hindered carbon, yielding the **less substituted alkene** as the major product.

2. Organometallic Reagents & Carbon-Carbon Bond Formation

1. **Grignard Reagents (Organomagnesium Halides, $R - MgX$):** Discovered by Victor Grignard, formed by reacting alkyl or aryl halides with magnesium metal in anhydrous "dry ether":

$$R - X + Mg \xrightarrow{\text{dry ether}} R - MgX$$

   - **Bond Polarity:** Carbon-magnesium bond is highly covalent but strongly polarised ($R^{\delta-} - Mg^{\delta+} X$), making carbon act as a powerful nucleophilic carbanion (R^-).
   - **Protonolysis with Active Hydrogen:** Grignard reagents react instantaneously with any compound containing active hydrogen (water, "alc"ohols, amines, terminal alkynes) to form alkanes:

$$R - MgX + H_2\ O \rightarrow R - H + Mg(OH)X$$

$$R - MgX + D_2\ O \rightarrow R - D + Mg(OD)X$$

   Crucial Experimental Rule: Grignard syntheses must be carried out in strictly anhydrous conditions; even trace atmospheric moisture destroys the reagent.

2. **Wurtz Reaction:** Two moles of alkyl halide are heated with sodium metal in "dry ether" to yield a symmetrical alkane containing double the carbon atoms:

$$2R - X + 2\ Na \xrightarrow{\text{dry ether}} R - R + 2\ NaX$$

   - **Limitation:** Unsuitable for preparing alkanes with an odd number of carbon atoms. Mixing two different alkyl halides ($R - X + R' - X$) gives an inseparable mixture of three alkanes ($R - R, R' - R', R - R'$) in poor yields. Methane cannot be synthesised.

3. **Fittig and Wurtz-Fittig Reactions:**
   - **Wurtz-Fittig Reaction:** A mixture of an alkyl halide and an aryl halide is treated with sodium in "dry ether" to yield an alkylarene:

$$C_6H_5\ Br + CH_3\ Br + 2\ Na \xrightarrow{\text{dry ether}} C_6H_5 - CH_3 + 2\ NaBr \quad \text{(Toluene)}$$

   - **Fittig Reaction:** Two moles of aryl halide react with sodium in "dry ether" to produce diphenyl / biphenyl:

$$2C_6H_5\ Cl + 2\ Na \xrightarrow{\text{dry ether}} C_6H_5 - C_6H_5 + 2\ NaCl \quad \text{(Biphenyl)}$$

- **Ullmann Reaction:** Heating iodobenzene with copper powder in a sealed tube yields biphenyl:

$$2C_6H_5 - I + Cu \xrightarrow{200^\circ\ C} C_6H_5 - C_6H_5 + CuI_2$$

Module 5: Chemical Reactivity of Haloarenes

1. Inertness of Haloarenes to Nucleophilic Substitution

Aryl halides are extraordinarily unreactive towards nucleophilic substitution compared to alkyl halides due to four reinforcing electronic and structural factors:

1. **Resonance Effect:** The lone pairs on the halogen atom delocalise into the aromatic π-electron system. This imparts **partial double-bond character** to the $C - X$ bond (169 pm in chlorobenzene versus 178 pm in chloroethane), making bond cleavage much harder.
2. **Difference in Hybridisation of Carbon:** In haloarenes, the carbon holding halogen is sp^2-hybridised (33% s-character), whereas in haloalkanes it is sp^3-hybridised (25% s-character). The greater s-character holds the bonding electrons tighter to the nucleus, shortening and strengthening the $C - X$ bond.
3. **Instability of Phenyl Cation:** Self-ionisation to form a phenyl cation ($C_6H_5^+$) is energetically prohibitive because the positive charge is in an sp^2 orbital orthogonal to the π-cloud and cannot be resonance-stabilised. This completely rules out the S_N1 pathway.
4. **Electrostatic Repulsion:** The electron-rich aromatic π-cloud repels the approaching nucleophile, preventing backside attack (S_N2).

2. Nucleophilic Aromatic Substitution (S_NAr) & Activation by Nitro Groups

Under drastic conditions, chlorobenzene undergoes substitution (Dow's Process):

$$C_6H_5\ Cl + NaOH \xrightarrow{623\ K, 300\ atm} C_6H_5\ ONa \xrightarrow{H^+} C_6H_5\ OH \quad \text{(Phenol)}$$

- **Activating Effect of Electron-Withdrawing Groups ($-NO_2$):** The introduction of electron-withdrawing nitro groups at the **ortho** and **para** positions drastically accelerates S_NAr by stabilizing the negative charge of the intermediate carbanion (**Meisenheimer complex**):
  - **4-Nitrochlorobenzene:** Hydrolyses with 15% NaOH at 443 K.
  - **2,4-Dinitrochlorobenzene:** Hydrolyses with aqueous Na_2CO_3 at 368 K.
  - **2,4,6-Trinitrochlorobenzene (Picryl chloride):** Hydrolyses simply upon warming with **pure water at 323 K** (50° C) to produce picric acid (2,4,6-trinitrophenol).
- **Why Meta-Nitro Exerts No Activation:** Resonance analysis demonstrates that the negative charge in the Meisenheimer intermediate is delocalised exclusively to the **ortho** and **para** ring carbons. A nitro group at the meta-position cannot directly stabilize the carbanionic charge via resonance; hence, meta-substituted haloarenes show negligible reactivity enhancement.

3. Electrophilic Substitution Reactions of Haloarenes

Halogen atoms on benzene exert two conflicting effects:

- **Inductive Effect ($-I$):** Halogen is strongly electronegative and withdraws electron density from the ring, **deactivating** the aromatic ring relative to benzene.

- **Resonance Effect ($+R$):** Delocalisation of lone pairs increases electron density specifically at the **ortho** and **para** positions.
- **Net Result:** Electrophilic substitution in haloarenes occurs slower than in benzene (due to overall deactivation), but proceeds exclusively at the **ortho** and **para** positions (**ortho/para directing**). The **para-isomer is always the major product** due to minimal steric hindrance.
  1. **Halogenation:** Chlorobenzene $+Cl_2 \xrightarrow{\text{anhy. } FeCl_3}$ 1,4-dichlorobenzene (major) + 1,2-dichlorobenzene (minor).
  2. **Nitration:** Chlorobenzene $+\frac{HNO_3}{H_2}SO_4 \rightarrow$ 1-chloro-4-nitrobenzene (major) + 1-chloro-2-nitrobenzene.
  3. **Sulphonation:** Chlorobenzene +conc. $H_2SO_4 \xrightarrow{\Delta}$ 4-chlorobenzenesulphonic acid (major).
  4. **Friedel-Crafts Alkylation:** Chlorobenzene $+CH_3\ Cl \xrightarrow{\text{anhy. } AlCl_3}$ 1-chloro-4-methylbenzene (major).
  5. **Friedel-Crafts Acylation:** Chlorobenzene $+CH_3\ COCl \xrightarrow{\text{anhy. } AlCl_3}$ 4-chloroacetophenone (major).

Module 6: Polyhalogen Compounds & Environmental Impact

1. **Dichloromethane (Methylene Chloride, CH_2Cl_2):**
   - Widely utilised as an industrial solvent, paint remover, aerosol propellant, and pharmaceutical process solvent.
   - **Toxicology:** Depresses the central nervous system (dizziness, nausea, numbness); direct contact causes skin burning and corneal burns.
2. **Trichloromethane (Chloroform, $CHCl_3$):**
   - Historically used as an inhalation anaesthetic; replaced today by safer agents due to liver and kidney toxicity.
   - **Oxidation to Lethal Phosgene:** In the presence of sunlight and atmospheric air, chloroform is slowly oxidized into carbonyl chloride (**phosgene**, $COCl_2$), a lethal chemical warfare gas:

$$2CHCl_3 + O_2 \xrightarrow{h\nu} 2COCl_2 + 2\ HCl$$

   - **Storage Protocol:** Chloroform is stored in completely filled, dark amber-coloured glass bottles stoppered tightly to exclude air and light. Additionally, 1% ethanol is added as a stabilizer: ethanol reacts with any formed phosgene to convert it into non-toxic diethyl carbonate:

$$COCl_2 + 2C_2H_5\ OH \rightarrow (C_2H_5\ O)_2\ CO + 2\ HCl$$

3. **Triiodomethane (Iodoform, CHI_3):**
   - A pale yellow crystalline solid with an unmistakable pungent medicinal odour ($m.p.$ 392 K).
   - Used historically as an antiseptic dressing for wounds. Its antiseptic property is not due to iodoform itself, but is entirely due to the **slow liberation of free iodine** when in contact with organic wound tissue.
4. **Tetrachloromethane (Carbon Tetrachloride, CCl_4):**
   - Manufactured on large scale as a feedstock for chlorofluorocarbons, dry-cleaning agent, and fire extinguisher (sold under commercial name **Pyrene**).
   - Highly toxic; exposure damages nerve cells and causes liver cancer. Discharged into the atmosphere, it migrates to the stratosphere, releasing chlorine radicals that destroy the ozone layer.

5. **Freons (Chlorofluorocarbons, CFCs):**
   - Extremely stable, non-flammable, non-toxic, and unreactive chlorofluoro derivatives of methane and ethane.
   - **Freon-12 (Dichlorodifluoromethane, CF_2Cl_2):** Manufactured via the Swarts reaction from CCl_4 and SbF_3. Extensively used as refrigerants and aerosol propellants.
   - **Ozone Depletion:** In the stratosphere, intense UV radiation breaks the $C-Cl$ bond, generating chlorine free radicals (Cl°) that catalytically destroy ozone molecules:

$$CF_2Cl_2 \xrightarrow{h\nu} CF_2Cl^{\circ} + Cl^{\circ}; \quad Cl^{\circ} + O_3 \rightarrow ClO^{\circ} + O_2; \quad ClO^{\circ} + O \rightarrow Cl^{\circ} + O_2$$

   A single chlorine radical destroys over $100,000$ ozone molecules. Governed internationally under the **Montreal Protocol**.

6. **DDT (p,p'-Dichlorodiphenyltrichloroethane):**
   - First chlorinated organic insecticide, synthesized by Paul Müller in 1939 (Nobel Prize 1948).
   - Prepared by condensing chlorobenzene with chloral ($CCl_3\ CHO$) in the presence of concentrated H_2SO_4:

$$2C_6H_5\ Cl + CCl_3\ CHO \xrightarrow{\text{conc. } H_2SO_4} (\text{p-Cl-}C_6H_4)_2\ CH - CCl_3 + H_2\ O$$

   - **Ecological Bioaccumulation:** Highly fat-soluble and chemically indestructible; cannot be metabolized by animals. Accumulates progressively up food chains (**biomagnification**), causing severe calcium metabolism disruption in birds (thin eggshells and population crashes). Banned globally.

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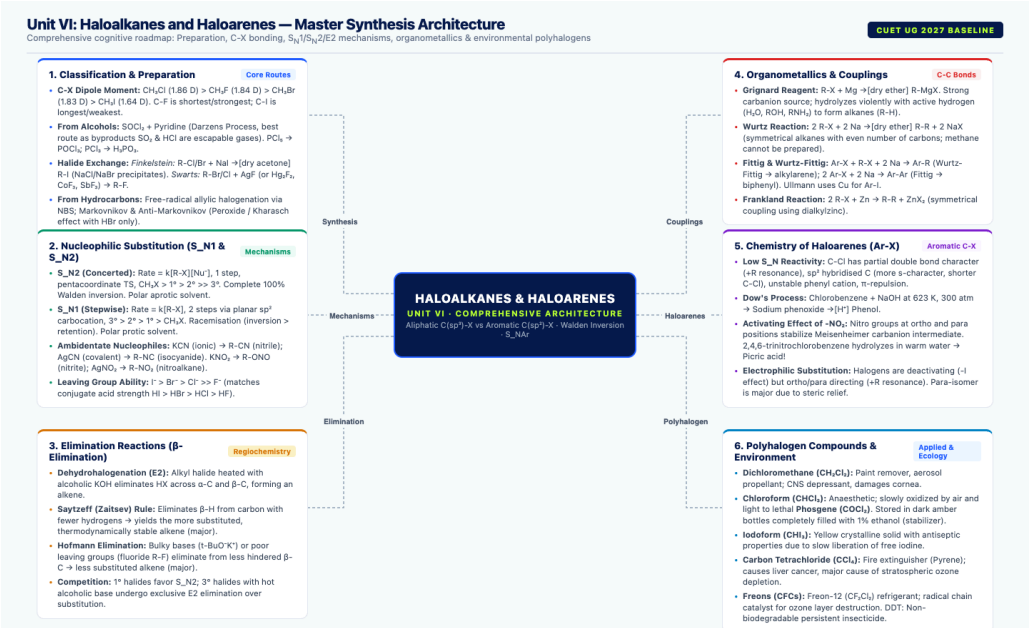


Figure 3: Master Synthesis Architecture: Complete Unit VI cognitive roadmap integrating preparation routes, C-X bond properties, substitution mechanisms, organometallics, and environmental polyhalogens.

UNIT VI MASTER FORMULA & HIGH-YIELD CHEATSHEET

Core Concept / Law	Expression / Formulation	Application & CUET Problem Strategy
Dipole Moment Order	CH_3 Cl (1.86 D) > CH_3F(1.85 D) > CH_3 Br > CH_3I	Longer $C-Cl$ bond lever arm beats higher electronegativity of fluorine.
Darzens Process	$R-OH + SOCl_2 \xrightarrow{\text{Pyridine}} R-Cl + SO_2\uparrow + HCl\uparrow$	Best route for pure alkyl chlorides; gaseous byproducts escape.
Finkelstein Reaction	$R-X + NaI \xrightarrow{\text{dry acetone}} R-I + NaX\downarrow$ $(X = Cl, Br)$	Driven by precipitation of insoluble $\frac{NaCl}{NaBr}$ in "dry acetone".
Swarts Reaction	$R-X + \frac{AgF}{Hg_2}\frac{F_2}{CoF_3} \rightarrow R-F + AgX\downarrow$	Primary commercial/lab method to synthesise volatile fluoroalkanes.
S_N2 Substitution	Rate $= k[R-X][N^-]$; 1 step, 100% Walden Inversion	Reactivity: $CH_3X > 1^\circ > 2^\circ \gg 3^\circ$. Favoured by strong Nu and polar aprotic solvents.
S_N1 Substitution	Rate $= k[R-X]$; 2 steps via planar R^+, Racemisation	Reactivity: $3^\circ > 2^\circ > 1^\circ$. Favoured by weak Nu, polar protic solvents, and allylic/benzylic halides.
Ambidentate Cyanide	$KCN \rightarrow R-CN$ (nitrile); $AgCN \rightarrow R-NC$ (isocyanide)	Ionic KCN attacks via C; covalent AgCN coordinates via N lone pair.
Saytzeff Elimination	Alkyl halide + alc. KOH → More substituted alkene	Major product has maximum alkyl groups attached to the double bond.
Grignard Reagents	$R-X + Mg \xrightarrow{\text{dry ether}} R-MgX \xrightarrow{H_2O} R-H$	Strong carbanion (R^-); must be prepared in strictly anhydrous "dry ether".
Wurtz / Fittig Couplings	$2R-X \rightarrow R-R$; $2\ Ar-X \rightarrow$ Ar-Ar; $Ar-X + R-X \rightarrow Ar-R$	Coupling via sodium in "dry ether". Wurtz synthesises even-carbon alkanes.
Dow's Process	C_6H_5 Cl + NaOH $\xrightarrow{623\ K, 300\ atm}$ C_6H_5 ONa $\xrightarrow{H^+}$ C_6H_5 OH	Haloarenes resist substitution; require extreme heat/pressure or ortho/para $-NO_2$.
Haloarene Directing	Halogen is deactivating $(-I > +R)$, but ortho/para directing	Electrophilic substitution is slower than benzene; para-isomer is major.
Phosgene Formation	$2CHCl_3 + O_2 \xrightarrow{h\nu} 2COCl_2 + 2\ HCl$	Prevented by storage in dark amber full bottles with 1% ethanol stabilizer.

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. Which of the following alkyl halides undergoes nucleophilic substitution by the S_N2 mechanism at the FASTEST rate?

(A) $(CH_3)_3$ C-Br **(B)** $(CH_3)_2$ CH-Br
(C) $CH_3CH_2CH_2$ Br **(D)** CH_3 Br

Sol. Q1. S_N2 proceeds via a concerted backside attack in a single step. The reaction rate is strictly governed by steric hindrance around the α-carbon. Unhindered methyl bromide (CH_3 Br) provides completely unobstructed backside access, reacting fastest: $CH_3X > 1^\circ > 2^\circ \gg 3^\circ$. **Answer: (D)**

Q2. When optically active $(-)-2-$bromooctane is reacted with aqueous sodium hydroxide under typical S_N2 conditions, the configuration of the resulting $2-$octanol exhibits:

(A) Complete retention of configuration **(B)** Complete inversion of configuration (Walden Inversion)
(C) A 50:50 racemic mixture of enantiomers **(D)** Partial racemisation with net retention of configuration

Sol. Q2. In an S_N2 substitution, the incoming nucleophile (OH^-) attacks the substrate strictly from the side opposite (180°) to the leaving bromide ion. This backside displacement forces all three non-reacting groups to invert spatial orientation like an umbrella in a storm (**Walden Inversion**), yielding pure $(+)-2-$octanol. **Answer: (B)**

Q3. The correct decreasing order of reactivity towards S_N1 hydrolysis among the following isomeric butyl bromides is:

(A) 1-bromobutane > 2-bromobutane > tert-butyl bromide **(B)** tert-butyl bromide > 2-bromobutane > isobutyl bromide > 1-bromobutane
(C) 1-bromobutane > isobutyl bromide > 2-bromobutane > tert-butyl bromide **(D)** tert-butyl bromide > 1-bromobutane > 2-bromobutane > isobutyl bromide

Sol. Q3. S_N1 reaction rate is determined by the ionization step forming a carbocation intermediate. Carbocation stability follows $3^\circ > 2^\circ > 1^\circ$ due to hyperconjugation and inductive effects: $(CH_3)_3C^+(3^\circ) > CH_3CH_2CH^+CH_3(2^\circ) > (CH_3)_2CHCH_2^+(1^\circ) > CH_3CH_2CH_2CH_2^+(1^\circ)$. Hence, tert-butyl bromide reacts fastest. **Answer: (B)**

Q4. The cleanest and most preferred laboratory method for preparing an alkyl chloride from an "alc"ohol uses which of the following reagents?

(A) PCl_5 in anhydrous ether **(B)** Concentrated HCl with anhydrous $ZnCl_2$ (Lucas reagent)
(C) $SOCl_2$ in the presence of pyridine (Darzens Process) **(D)** Cl_2 gas irradiated under ultraviolet light

Sol. Q4. Reaction with thionyl chloride ($SOCl_2$) is the cleanest preparation because both byproduct compounds—sulfur dioxide (SO_2) and hydrogen chloride (HCl)—are gases that escape from the reaction mixture, leaving behind virtually pure alkyl chloride without necessitating tedious extraction or distillation. **Answer: (C)**

Q5. When ethyl bromide (CH_3CH_2 Br) is treated separately with potassium cyanide (KCN) and silver cyanide (AgCN), the major products obtained are respectively:

(A) CH_3CH_2 CN and CH_3CH_2 CN **(B)** CH_3CH_2 NC and CH_3CH_2 CN
(C) CH_3CH_2 CN and CH_3CH_2 NC **(D)** CH_3CH_2 NC and CH_3CH_2 NC

Sol. Q5. Cyanide is an ambidentate nucleophile. Potassium cyanide (KCN) is ionic and provides cyanide ions; because the $C-C$ bond is more stable than the $C-N$ bond, nucleophilic attack occurs through carbon, forming ethyl cyanide / propanenitrile (CH_3CH_2 CN). Silver cyanide (AgCN) is predominantly covalent; only the nitrogen atom possesses a free lone pair to donate, producing ethyl isocyanide (CH_3CH_2 NC). **Answer: (C)**

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. Heating 2-bromopentane with hot "alc"oholic potassium hydroxide (alc. KOH) produces pent-2-ene as the major product (≈ 81%) rather than pent-1-ene. This regiochemical outcome is governed by:

(A) Markovnikov's addition rule
(B) Hofmann's elimination rule
(C) Saytzeff's (Zaitsev's) elimination rule
(D) Kharasch peroxide effect

Sol. Q6. According to Saytzeff's rule, base-induced dehydrohalogenation of an unsymmetrical alkyl halide eliminates a β-hydrogen preferentially from the carbon atom bonded to fewer hydrogens, yielding the more substituted alkene as the major product because it possesses greater thermodynamic stability through hyperconjugative resonance. **Answer: (C)**

Q7. Despite the higher electronegativity of fluorine compared to chlorine, the experimental dipole moment of chloromethane (CH_3 Cl, 1.860 D) is greater than that of fluoromethane (CH_3F, 1.847 D). This anomaly occurs because:

(A) Fluorine possesses three non-bonding lone pairs that cancel its dipole moment
(B) The $C-$Cl bond length is significantly longer, compensating for the lower charge separation
(C) Fluoromethane exists as a symmetrical planar dimer in the gas phase
(D) The polarizability of the methyl group is completely neutralized by fluorine

Sol. Q7. Dipole moment (μ) is defined as the mathematical product of partial charge separation (q) and internuclear distance (d): $\mu = q \times d$. Although the charge separation in $C-F$ is slightly larger due to fluorine's electronegativity, the $C-$Cl bond length (178 pm) is substantially greater than the $C-F$ bond (139 pm). The longer distance factor outweighs the charge difference, making CH_3 Cl more polar. **Answer: (B)**

Q8. An alkyl halide $R-X$ is converted into a Grignard reagent $R-$MgX in "dry ether". When this reagent is treated with heavy water (D_2 O), the hydrocarbon product formed is:

(A) $R-$OH
(B) $R-$OD
(C) $R-$D
(D) $R-R$

Sol. Q8. The carbon-magnesium bond in a Grignard reagent is highly polarized ($R^{\delta-}-Mg^{\delta+}X$), behaving as an extraordinarily strong carbanionic base. It abstracts deuterium from deuterated water (D_2 O) instantly via protonolysis: $R-\text{MgX} + D_2\text{O} \rightarrow R-\text{D} + \text{Mg(OD)}X$. **Answer: (C)**

Q9. Which of the following haloarenes undergoes nucleophilic aromatic substitution (S_NAr) with aqueous sodium hydroxide under the MILDEST reaction conditions?

(A) Chlorobenzene
(B) 4-Chloronitrobenzene
(C) 2,4-Dinitrochlorobenzene
(D) 2,4,6-Trinitrochlorobenzene (Picryl chloride)

Sol. Q9. Electron-withdrawing nitro groups ($-NO_2$) at the ortho and para positions strongly stabilize the negative charge in the Meisenheimer intermediate via resonance. In 2,4,6-trinitrochlorobenzene, three $-NO_2$ groups activate the ring so intensely that simple warming with neutral water at 323 K (50° C) causes quantitative hydrolysis to picric acid. **Answer: (D)**

Q10. Chlorobenzene is extraordinarily resistant to nucleophilic substitution at room temperature compared to chloroethane. Which of the following factors does NOT contribute to this inertness?

(A) Delocalisation of chlorine lone pairs giving the $C-$Cl bond partial double-bond character
(B) The higher electronegativity of sp^2-hybridised ring carbon holding the bond tighter
(C) The extreme thermodynamic instability of the hypothetical phenyl cation intermediate
(D) The exceptionally high nucleophilicity of aromatic ring hydrogens

Sol. Q10. Aryl halide inertness arises from: (1) resonance partial double-bond character, (2) sp^2 carbon with 33% s-character shortening the bond, (3) instability of phenyl cation preventing S_N1, and (4) electrostatic repulsion between the π-cloud and incoming nucleophiles. The nucleophilicity of ring hydrogens is physically meaningless in this context. **Answer: (D)**

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. In the Finkelstein reaction, alkyl iodides are prepared by reacting alkyl bromides or chlorides with sodium iodide in "dry acetone". The principal driving force shifting the reaction to completion is:

(A) The high volatility of alkyl iodides evaporating from the solvent
(B) The precipitation of sodium chloride or sodium bromide in "dry acetone"
(C) The strong coordination of "dry acetone" with alkyl carbocations
(D) The instantaneous decomposition of hydrogen iodide by-product

Sol. Q11. Sodium iodide (NaI) is uniquely soluble in "dry acetone", whereas sodium chloride (NaCl) and sodium bromide (NaBr) are insoluble covalent precipitates in this organic medium. As NaX precipitates out of solution, Le Chatelier's principle shifts the equilibrium forward towards the formation of pure alkyl iodide. **Answer: (B)**

Q12. A mixture of bromobenzene and 1-bromobutane is treated with sodium metal in "dry ether". The major alkylarene obtained from the cross-coupling reaction is:

(A) Biphenyl
(B) Octane
(C) Butylbenzene
(D) 1,2-Diphenylbutane

Sol. Q12. This cross-coupling between an aryl halide and an alkyl halide using sodium metal in "dry ether" is the **Wurtz-Fittig reaction**: $C_6H_5\ Br + CH_3CH_2CH_2CH_2\ Br + 2\ Na \xrightarrow{\text{dry ether}} C_6H_5 - CH_2CH_2CH_2CH_3 + 2\ NaBr$. The principal product is butylbenzene. **Answer: (C)**

Q13. Which of the following halogenated compounds possesses a chiral carbon centre and is therefore capable of exhibiting optical isomerism?

(A) 1-Chlorobutane
(B) 2-Chlorobutane
(C) 1-Chloro-2-methylpropane
(D) 2-Chloro-2-methylpropane

Sol. Q13. A carbon atom is chiral (asymmetric) when bonded to four entirely different atoms or groups. In 2-chlorobutane, C2 is bonded to: (1) a hydrogen atom ($-H$), (2) a chlorine atom ($-Cl$), (3) a methyl group ($-CH_3$), and (4) an ethyl group ($-CH_2CH_3$). Because all four ligands are non-identical, it lacks symmetry and exists as optical enantiomers. **Answer: (B)**

Q14. Chloroform ($CHCl_3$) is stored in completely filled dark amber-coloured bottles containing approximately 1% pure ethanol. The vital role of ethanol in this mixture is to:

(A) Prevent the evaporation of chloroform into the atmosphere
(B) Act as a reducing agent that converts deadly phosgene into non-toxic diethyl carbonate
(C) Accelerate the conversion of chloroform into carbon tetrachloride
(D) Function as a colour indicator for atmospheric moisture leakage

Sol. Q14. Chloroform slowly oxidizes in air and light to form carbonyl chloride (**phosgene**, $COCl_2$), a lethal poison. Adding 1% ethanol serves a dual safety purpose: it suppresses oxidation, and any trace phosgene formed reacts with ethanol to produce non-toxic diethyl carbonate: $COCl_2 + 2C_2H_5\ OH \rightarrow (C_2H_5\ O)_2\ CO + 2\ HCl$. **Answer: (B)**

Q15. The synthetic insecticide DDT (p,p'-dichlorodiphenyltrichloroethane) was banned globally primarily because:

(A) It hydrolyses instantly upon spraying, losing all insecticidal potency
(B) It is highly volatile and produces dense toxic smogs in urban centers
(C) It is chemically persistent, fat-soluble, and biomagnifies across ecological food chains
(D) It reacts with stratospheric ozone via free radical chlorination

Sol. Q15. DDT is extremely non-biodegradable and fat-soluble. When ingested by insects, aquatic organisms, and fish, it cannot be metabolized or excreted, depositing in lipid tissues. Its concentration multiplies exponentially at successive trophic levels (**biomagnification**), causing critical calcium metabolic collapse and reproductive failure in apex predators. **Answer: (C)**

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